AHMED FATHY GWELY

AI Engineer |Deep Learning · Edge AI & IoT

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PROFESSIONAL SUMMARY

AI Engineer specializing in computer vision, deep learning, and real-time AI systems for IoT and edge deployment. Deployed an end-to-end video and sensor pipeline to production for water-station automation (Hyper SCADA) and built YOLO-based detection models running on a Raspberry Pi 5 for **GallineX**, a poultry-monitoring system that began as a graduation project and is now being commercialized with research and industry funding. Experienced across the full ML lifecycle — labeling, training, evaluation, ONNX deployment, live monitoring — and in integrating models with ESP32 and Firebase. B.Sc. Electrical Engineering; trained and certified in LLM application development .

TECHNICAL SKILLS

Programming: Python, NumPy, asynchronous Python
Machine Learning & Deep Learning: CNNs, GANs, object detection, segmentation, model training & evaluation, PyTorch (Ultralytics)
Computer Vision: YOLO (Ultralytics), OpenCV, MediaPipe, object detection & tracking, pose estimation, image processing, dataset annotation (Roboflow), data augmentation
Data Science: Statistical modeling, feature engineering, data cleaning & preprocessing, data visualization
AI Deployment & MLOps: ONNX / ONNX Runtime, end-to-end pipelines (ingestion → inference → messaging), production monitoring
IoT & Edge AI: Raspberry Pi 5, ESP32, embedded Linux, MQTT, Firebase Realtime Database, wireless protocols, on-device inference
Generative AI: Prompt engineering, RAG pipelines (training & certification), tool-based / agentic reasoning pipelines
Tools & Frameworks: Git, Streamlit, Roboflow, Kaggle (GPU training), Power BI

PROFESSIONAL EXPERIENCE

AI Engineer Facilitator 07/2026 – Present
Samsung Innovation Campus (SIC), with Samsung Electronics Egypt & Life Makers — Cairo

- Deliver applied AI and machine learning training sessions to university students, covering data preprocessing, model building, and evaluation, as part of the SIC national program.

Deep Learning Engineer (Intern) | *Lotus Power Company — Cairo & Vancouver* 12/2025 – 2026/8

- Develop computer vision and machine learning models for IoT-based smart farm management, combining visual detection with sensor data to drive automated decisions.
- Integrate model outputs into real-time farm monitoring with reliable on-device inference, working with embedded systems, power electronics, and software teams.
- Contribute to research, technical documentation, and patent/publication material for the agricultural technology.

Deep Learning Engineer (Computer Vision & IoT), Freelance 04/2024 – 11/2025
Hyper SCADA — Egyptian Joint-Stock Company, Water Sector

- Designed and deployed to production an end-to-end pipeline (video/sensor ingestion → preprocessing → model inference → messaging) for real-time water-station automation, turning raw sensor and video data into operational decisions.
- Owned the full model lifecycle: custom dataset collection and labeling, training, evaluation, deployment, and live monitoring.
- Iterated on the deployed model to reduce human intervention and operational costs; shipped updates via Git.

IoT Engineer (Intern) | *Samsung Innovation Campus — Smart City Track* 08/2025

- Worked on embedded Linux and IoT data pipelines supporting smart city applications.

Undergraduate Researcher | *Zewail City Computing Society (ZCCS)* 11/2024

- Conducted undergraduate research in applied artificial intelligence and machine learning.

AI Engineer (Intern) | *Samsung Innovation Campus — Giza* 07/2024 – 10/2024

- Ranked top 2 on the leaderboard in internal machine learning and deep learning modeling competitions.

PROJECTS

GallineX — Smart Poultry Farm Monitoring & Power Management System 2025 – 2026
Graduation Project · 11-member team · Funded by Lotus Power Company

- **Commercialization & funding:** Developed the graduation project into a final product — **sales have started**. Accepted for funding by the **Academy of Scientific Research and Technology (ASRT)**; finalist in the **Egypt IoT & AI Challenge**, with financial support from the **Information Technology Industry Development Agency (ITIDA)**.
- **System:** AI monitoring, IoT, and power electronics (PFC + LLC resonant AC charging, solar MPPT, 48V battery) that keep poultry farms powered and monitored where grid supply is unreliable. Contributed the AI and computer-vision subsystem within a multidisciplinary electrical engineering team.
- **Behavior recognition:** Trained a YOLO26m model on annotated frames from real farm video for bird detection, counting, tracking, and classification of eating/drinking, moving, and resting behavior in real time.

- **Health detection:** Trained a YOLO11n model to flag healthy vs. unhealthy birds, with an analytics layer that estimates the unhealthy-bird percentage and generates alerts and reports.
- **Training & deployment:** Trained with Ultralytics YOLO on Kaggle GPUs with data augmentation and evaluated both models on held-out test sets; exported them to ONNX and deployed on a desktop app, a Streamlit web app, and an offline **Raspberry Pi 5** edge device processing live camera video. Integrated with ESP32 sensors and Firebase for real-time sync and remote monitoring.

  Tech stack: Python, Ultralytics YOLO (YOLO26m, YOLO11n), PyTorch, ONNX Runtime, OpenCV, Streamlit, Raspberry Pi 5, ESP32, Firebase, Roboflow, Kaggle

Other Projects

- **Smart Emotion-Detecting Sunglasses** — Real-time emotion-classification model running on a Raspberry Pi wearable, with audio feedback for blind users.
- **Breast Cancer Detection** — CNN + GAN model for tumor detection and classification from medical imaging data. **Brain Tumor Segmentation** — Deep learning pipeline for automatic MRI-based tumor detection and segmentation.
- **Azmuth AI Assistant (GitHub)** — Local-first AI assistant with tool-based reasoning, persistent memory, and an async multi-threaded architecture for OS-level task automation.
- **Neural Network from Scratch** — Digit-recognition network implemented with NumPy only (backpropagation and optimization, no ML framework).
- **Additional computer vision & generative projects** — Object and vehicle detection, garbage detection, pose estimation, sign language detection, GAN-based anime face generation, neural style transfer (Portfolio / GitHub / Kaggle).

SPEAKING, TEACHING & LEADERSHIP

Conference Speaker — Opening Session | *First Student Engineering Conference on AI Applications — Mansoura University* 01/2026

- Delivered "AI Evolution: Narrow to General Intelligence and Beyond" in the opening session — the only student selected to present, following the Head of the CCE Department and the Vice Dean for Education and Student Affairs; covered ANI versus AGI, LLM and computer-vision limits, computational neuroscience and model scaling, generative AI and physical AI.

AI & Computer Vision Instructor | *Neurobotics* 01/2025 – Present

- Instructor and technical program lead for a computer vision and applied machine learning curriculum.

Deep Learning Supervisor / Instructor / Mentor | *CAT Reloaded, Mansoura University* 03/2025 – 11/2025

- Supervised and mentored students in deep learning; delivered the Computer Vision workshop "How to Make Machines See?" in 2025 and 2026.

Deep Learning Supervisor | *IEEE Mansoura Computer Society — Data Science Committee* 01/2025 – 10/2025

- Mentored students and led instructional sessions on deep learning and data science fundamentals.

Head of AI Committee | *Luminous MU* 06/2024 – 09/2025

- Led the AI committee and its technical education programme; previously AI / Computer Vision Instructor (10/2023 – 05/2025).

Head of AI Committee & Co-Founder | *Momentum MU* 02/2023 – 09/2025

- Co-founded and led the AI Committee; served as Vice Head of Training (03/2023 – 08/2023) and as a public speaker for data science and AI initiatives.

EDUCATION

B.Sc. Electrical Engineering | *Mansoura University* 10/2021 – 06/2026

- Overall Grade: Very Good.
- Graduation Project: Excellent — GallineX, a funded and market-launched smart power and monitoring system.

CERTIFICATIONS & TRAINING

Professional Certifications

- Oracle Cloud Infrastructure (OCI) 2025 Certified Generative AI Professional
- Oracle Cloud Infrastructure (OCI) 2025 Certified Data Science Professional

Training Programs & Courses

- Samsung Innovation Campus — AI Engineer Program (2024); IoT Engineer Program (2025)
- Information Technology Institute (ITI) — NVIDIA DLI Generative AI Summer Training Program (09/2025 – 10/2025)
- NVIDIA Deep Learning Institute — Building RAG Agents with LLMs; Building LLM Applications with Prompt Engineering
- Udacity — Introducing Generative AI with AWS (Amazon Bedrock, SageMaker, CodeWhisperer)
- IBM — Data Analysis with Python; Applied Data Science Capstone
- Stanford & DeepLearning.AI — Supervised Machine Learning: Regression and Classification
- Self-directed — Computer Vision (OpenCV, MediaPipe, YOLOv8, CNNs)

COMPETITIONS & AWARDS

- **Finalist** — Egypt IoT & AI Challenge (GallineX), with financial support from ITIDA
- **2nd Place** — Industry 4.0 Transformation Marathon (20,000 EGP), 04/2024
- IEEE VICTORIS 2.0 — Micro Mouse Maze Solver, 08/2023; Smart Environment Hackathon — AI-powered sustainability solutions